

XIAOFENG TAN

Shenzhen, Guangdong, China

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RESEARCH INTERESTS

RL, AIGC, and world models, with a focus on human-centric perception, generation, and understanding.

EDUCATION

Southeast University (SEU), Nanjing, China

Sept 2024 – Present

CSRankings (China, 2016–2026): AI #16 · CV #25 · ARWU 2026 #89

M.Sc. in Computer Science [PALM Lab](#), supervised by Prof. [Hongsong Wang](#)

GPA: 88.6/100

First-Class Academic Scholarship (top 10%) & Tencent-SEU Scholarship (top 2.5%); Summer Camp Rank 2

Shenzhen University (SZU), Shenzhen, China

Sept 2020 – Jun 2024

CSRankings (China, 2016–2026): AI #21 · CV #20 · ARWU 2026 151–200

B.S. in Computer Science & B.S. in Mathematics (*Dual Degree, Honor*)

GPA: 88.9/100 · 4.07/4.50

Outstanding Graduate Representative (1/300+) and graduation speaker; featured in [SZU's 2024 Admissions Guide](#)

SELECTED PUBLICATIONS

* indicates equal contribution

◆ When Policy Entropy Constraint Fails: Preserving Diversity in Flow-based RLHF via Perceptual Entropy

Xiaofeng Tan, Jun Liu, Bin-Bin Gao, Yuanting Fan, Xi Jiang, Chengjie Wang, Hongsong Wang, Feng Zheng

[NeurIPS'26](#) · *Advances in Neural Information Processing Systems* (CCF-A, CORE-A*)

[Spotlight](#) (292/30,709 submissions, 0.95%)

[\[arXiv\]](#) [\[Project\]](#)

◆ MotionRFT: Unified Reinforcement Fine-Tuning for Text-to-Motion Generation

Xiaofeng Tan*, Wanjiang Weng*, Hongsong Wang, Fang Zhao, Xin Geng, Liang Wang

[TPAMI'26](#) · *IEEE Transactions on Pattern Analysis and Machine Intelligence* (CCF-A, SCI-Q1)

Extended from EasyTune (ICLR'26).

[\[arXiv\]](#) [\[Project\]](#) [\[Code\]](#)

EasyTune: Efficient Step-Aware Fine-Tuning for Diffusion-Based Motion Generation

Xiaofeng Tan*, Wanjiang Weng*, Haodong Lei, Hongsong Wang

[ICLR'26](#) · *International Conference on Learning Representations* (CCF-A, CORE-A*)

[\[PDF\]](#) [\[Project\]](#) [\[Openreview\]](#)

ReAlign: Text-to-Motion Generation via Step-Aware Reward-Guided Alignment

Wanjiang Weng*, Xiaofeng Tan*, Junbo Wang, Guo-Sen Xie, Pan Zhou, Hongsong Wang

[AAAI'26](#) · *AAAI Conference on Artificial Intelligence* (CCF-A, CORE-A*)

[\[PDF\]](#) [\[Project\]](#) [\[Code\]](#)

FG-Diff: Frequency-Guided Diffusion Model with Perturbation Training for Skeleton-Based VAD

Xiaofeng Tan, Hongsong Wang, Xin Geng, Liang Wang

[TIP'26](#) · *IEEE Transactions on Image Processing* (CCF-A, SCI-Q1)

[\[Project\]](#) [\[PDF\]](#) [\[Code\]](#)

SoPo: Text-to-Motion Generation Using Semi-Online Preference Optimization

Xiaofeng Tan, Hongsong Wang, Xin Geng, Pan Zhou

[NeurIPS'25](#) · *Advances in Neural Information Processing Systems* (CCF-A, CORE-A*)

[\[Project\]](#) [\[PDF\]](#) [\[Code\]](#)

Fuzzy Granule Density-Based Outlier Detection with Multi-Scale Granular Balls

Can Gao (Supervisor), Xiaofeng Tan, Jie Zhou, Weiping Ding, Witold Pedrycz

[TKDE'25](#) · *IEEE Transactions on Knowledge and Data Engineering* (CCF-A, SCI-Q1)

[\[PDF\]](#) [\[Code\]](#)

Dual-Grained Policy Optimization for Hallucination Reduction in Flow-based RL

Xiaofeng Tan, Jun Liu, Yuanting Fan, Bin-Bin Gao, Xi Jiang, Xiaochen Chen, Jinlong Peng, Chengjie Wang, Hongsong Wang, Feng Zheng

Under Review

[\[Project\]](#) [\[arXiv\]](#) [\[Code\]](#)

RESEARCH EXPERIENCE

Tencent Inc. , Shenzhen, China Research Intern @ HY Spatial reward modeling and curriculum-guided RL for text-to-image generation: tool-grounded spatial verification with SpatialCritic and rollouts progressing from foundational skills to complex compositions (Getting Space Right, under review; joint work with UC Merced).	Mar 2026 – Present <i>Mentor:</i> Jiaxin Lin
University of California, Merced , Merced, USA (Remote) Remote Visiting Student @ VLLab Remote collaboration on spatial reasoning and reward reliability for image generation, through the joint Getting Space Right project with Tencent HY.	Apr 2026 – Present <i>Mentor:</i> Ming-Hsuan Yang
Tencent Inc. , Shenzhen, China Research Intern @ YouTu Lab - Analyzed why policy entropy can remain constant while perceptual diversity collapses in flow-based RLHF; developed perceptual-entropy regularization to improve the quality–diversity trade-off (PEC, NeurIPS’26 Spotlight). - Investigated visual hallucinations caused by limited exploration and trajectory imitation; developed dynamic-granularity rollouts and consistent policy-gradient optimization (Consistent-RFT, under review).	Jul 2025 – Mar 2026 <i>Mentor:</i> Jun Liu
Singapore Management University , Singapore (Remote) Remote Visiting Student @ Language and Vision Lab (LV Lab) - Developed semi-online preference optimization that pairs online unpreferred motions with offline preferred examples, addressing offline overfitting and online sampling bias (SoPo, NeurIPS’25). - Co-developed step-aware reward modeling and reward-guided sampling to improve text-motion alignment and motion realism during denoising (ReAlign, AAAI’26; co-first author).	Jul 2024 – Mar 2025 <i>Mentor:</i> Pan Zhou
Southeast University , Nanjing, China Master’s Student @ PALM Lab - Co-developed step-wise reward optimization and a unified reward model for heterogeneous motion representations (EasyTune, ICLR’26; MotionRFT, TPAMI’26; co-first author on both). On HumanML3D with MLD, EasyTune achieves FID 0.132 at 22.10 GB peak memory, using 15.22 GB less than DRaFT-50. - Developed frequency-guided diffusion with perturbation training for skeleton-based video anomaly detection, improving robustness to unseen normal motion and emphasizing global-motion reconstruction (FG-Diff, TIP’26).	Apr 2024 – Present <i>Mentor:</i> Hongsong Wang
Shenzhen University , Shenzhen, China Undergraduate Research Assistant @ Computer Vision Institute Developed multi-scale granular-ball and fuzzy-rough-set methods for detecting heterogeneous outliers (MGBOD, TKDE’25; first student author), and three-way decision-based multi-view co-detection (TWCOD, IJAR’23; first author).	Oct 2022 – Oct 2023 <i>Mentor:</i> Can Gao

SERVICE

Conference Reviewer: ICML 2026 (*Gold Reviewer*), NeurIPS 2026 (*Top Reviewer*), AAAI 2027, ICLR 2027
Journal Reviewer: Transactions on Machine Learning Research (TMLR), International Journal of Data Science and Analytics (IJDSA)

HONORS & AWARDS

Master’s Studies		
ICML Gold Reviewer	International Conference on Machine Learning	2026
NeurIPS Top Reviewer	Conference on Neural Information Processing Systems	2026
Tencent-SEU Scholarship (top 2.5%)	Southeast University (¥9,000)	2025
First-Class Academic Scholarship (top 10%)	Southeast University (¥12,000)	2025
Undergraduate Studies		
Outstanding Graduate Representative	Shenzhen University (Graduation Speech)	2024
Featured in Admissions Guide	Shenzhen University (as Outstanding Student)	2024
Honor Bachelor Degree (top 3%)	Shenzhen University	2024
Outstanding Graduate (top 5%)	Shenzhen University	2024
LiYuan Star (SZU Special Award, top 0.73%)	Shenzhen University (¥20,000)	2022, 2023
Scholarship of Outstanding Innovative Talent	Guangdong Science College Entrance Exam (top 2%, ¥20,000)	2020